
Comprehensive Review on Vietnamese Speech Recognition Baselines

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Abstract

Automatic Speech Recognition task for the Vietnamese language remains a significant challenge due to complex tonal systems and high regional dialect variability. This study presents a comprehensive benchmark of five prominent ASR architectures Whisper Family, SqueezeFormer, Wav2Vec 2.0, and Qwen ASR evaluated across the dialect-diverse ViMD dataset and the standardized VIVOS corpus. Our evaluation encompasses both zero-shot and fine-tuned configurations, focusing on Word Error Rate and inference latency as primary metrics. Experimental results demonstrate that Qwen ASR achieves superior zero-shot generalization with a benchmark WER of 0.15 on the ViMD dataset. Conversely, fine-tuned Wav2Vec 2.0 provides the most optimal balance between accuracy and efficiency, matching top-tier performance with a peak inference speed of 0.5 s/sample. While Whisper variants exhibit consistent improvements through scaling, they are constrained by substantial computational overhead. These findings offer a strategic roadmap for deploying ASR systems tailored to the specific linguistic and operational requirements of Vietnamese speech applications

1 Introduction

Despite significant advances in the field of automatic speech recognition (ASR), driven by powerful deep learning architectures such as Whisper, Wav2Vec 2.0, SqueezeFormer and Qwen ASR [11][1][6][13], their application to Vietnamese is still limited.

Recent studies have shown that large-scale pre-training improves robustness and generalization across domains [1][6][11]. However, for low-resource but complex languages like Vietnamese, performance still depends heavily on the availability of monitored data to capture real-world variation, especially regional dialects. Existing datasets often present a trade-off between recording quality and linguistic diversity [10][2], making it difficult to develop models that generalize well across domains.

2 Related Work

2.1 Vietnamese ASR Datasets

Vietnamese automatic speech recognition datasets are generally categorized into two main types, including read speech and spontaneous speech, and each type has its own advantages and limitations.

Read speech data, including Common Voice (Vi) [9] and VIVOS [8], is generally collected in a controlled environment, in which the speakers are asked to speak certain texts. Read speech data is best suited for evaluating the performance of automated speech recognition systems in an environment with good acoustic conditions, as the data is of good quality with minimal noise. However, the data may not be simulated of the actual environment, as the data may not contain the variety of natural speech, which may contain various speaking styles, dialects, and noise.

On the other hand, spontaneous or natural speech datasets tend to include more varied acoustic conditions. For example, ViMD [3] datasets include speech collected from Vietnamese radio and television programs, representing a wide range of dialects across all 63 provinces. Besides varying linguistics, there is also varied acoustic noise and recording conditions. However, noise interference and varied speech patterns also come with this as a challenge.

The balance between clean speech and realistic, noisy speech is one of the main reasons why robust ASR systems are being developed, especially for low-resource languages with varied dialects like Vietnamese.

2.2 Pretrained and Vietnamese ASR Models

Pre-trained models based on self-supervised and weakly supervised learning have emerged as the leading paradigm in modern ASR systems. These models learn representations of speech data from large-scale corpora and can be used to solve downstream tasks via fine-tuning.

These models have been adapted to Vietnamese by fine-tuning them on Vietnamese corpora. Early works, such as Vietnamese ASR systems based on wav2vec2, demonstrate the effectiveness of pre-trained models in end-to-end ASR. Recent works, such as PhoWhisper [7], improve ASR performance by fine-tuning pre-trained models on a variety of Vietnamese speech corpora with different accents and speaking conditions.

In addition, some recent works aim to improve the efficiency of ASR systems and incorporate them with large-scale language models to improve performance.

However, most of the recent works on ASR systems were designed to improve the accuracy of the ASR systems on particular corpora, with less concern about comparisons between models and robustness to different conditions.

2.3 ASR under Domain Shift

While achieving excellent performance on benchmark sets, ASR models have been observed to degrade substantially on other data distributions. This phenomenon is called domain shift. Domain shift is usually caused by variations in acoustic environments, recording environments, speaking styles, and dialects. Previous works have demonstrated that ASR systems trained on clean and read speech may not perform optimally on noisy and spontaneous speech. However, most existing works were carried out in matched conditions. In the case of Vietnamese ASR, the domain shift is an important factor because of the tonal and dialectal characteristics of the language. In recent years, PhoWhisper [7] is reported to have greatly improved the robustness of the ASR model with the use of large and diverse training data. However, the effect of domain shift is not explicitly examined in an experimental setup. In addition, few works have directly compared different ASR architectures with the same domain shift.

3 Datasets

In this study, we used the ViMD [3] dataset collected from Vietnamese radio and television programs. ViMD [3] records the linguistic diversity of all 63 provincial dialects across Vietnam, as illustrated in the Figure 1; this regional diversity makes it a powerful resource for Vietnamese automatic speech recognition (ASR) tasks. The dataset contains over 100 hours of audio, including approximately 19,000 sentences and 1.2 million words in the recordings, which were used for fine-tuning.

To evaluate the performance of the baseline models across different acoustic environments, the VIVOS [8] corpus was employed for additional validation. Published by AILAB (VNUHCM - University of Science), this corpus consists of 15 hours of speech recorded by 65 speakers, totaling more than 12,000 utterances, detailed in Table 1. In contrast to the diverse, real-world conditions of ViMD [3], VIVOS [8] was recorded in controlled, quiet environments using high-quality microphones, with speakers providing clean, read-speech samples.

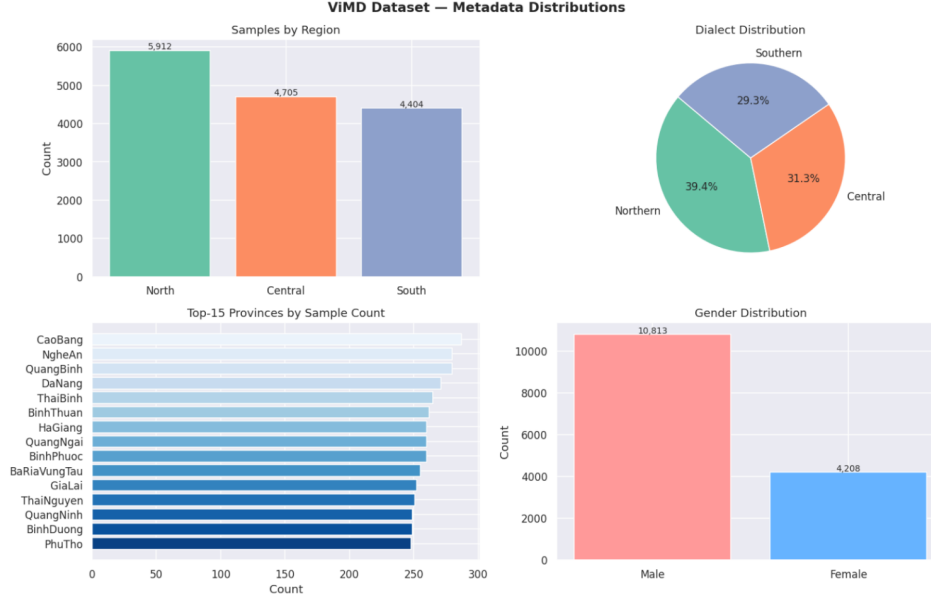


Figure 1: ViMD: Metadata Distribution

Statistic	Train	Test
Speakers	46	19
Utterances	11660	760
Duration	14:55	00:45
Unique Syllables	4617	1692

Table 1: VIVOS: Dataset Statistics

4 Method

4.1 Baseline Architectures

4.1.1 Whisper

Whisper [11], developed by OpenAI and released in 2022, is a robust automatic speech recognition (ASR) system trained on 680,000 hours of multilingual audio data. Pretrained checkpoints include variants such as tiny, base, small, medium, large-v2, and large-v3 with the parameter sizes range from 39 million (tiny) to 1.55 billion (large). Its architecture is an encoder-decoder transformer[14], which employs log-Mel spectrogram features extracted from 16 kHz audio, the model workflow is shown in Figure 2. The training objective is end-to-end sequence-to-sequence prediction, encompassing transcription, translation, language identification, and timestamping. In this research, we mainly focus on the Tiny, Base, and Small versions of Whisper [11] in order to compare their performance across different model scales.

As whisper model’s input accept the audio maximum duration is 30 seconds , preprocessing is required to ensure dataset compatibility while preserving transcription accuracy.

In the ViMD dataset [3], some audio recordings exceed the 30-second input limit required by Whisper [11]. Although Whisper [11] automatically pads shorter audio segments and trims longer ones, such truncation does not guarantee correct alignment with the corresponding transcripts. Misalignment between audio and text can negatively affect the performance of speech recognition models in training.

To address this issue, we developed a custom preprocessing tool that allows manual segmentation of long audio recordings. After applying this tool, longer recordings are divided into 2 shorter segments, with transcripts adjusted accordingly. We also manually verified each segment to ensure audio match with its corresponding transcript.

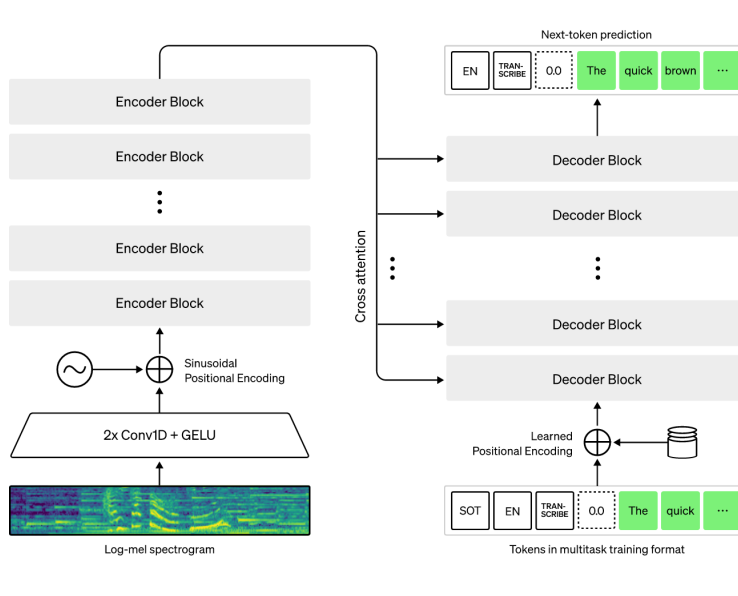


Figure 2: Illustration of Whisper model. Source: OpenAI

4.1.2 SqueezeFormer

The next architecture used in this study is SqueezeFormer [6], a Transformer-based encoder architecture [14]. Developed by researchers at UC Berkeley and collaborators and released in 2022, SqueezeFormer [6] improves upon Conformer [5] by incorporating a squeeze-and-expand strategy through a Temporal U-Net [12] structure, which reduces computational costs while preserving strong representational power.

Before being processed by the model, the input audio signal is first resampled to 16 kHz audio waveforms. The feature extraction process involves the following steps:

1. Short-Time Fourier Transform (STFT) to obtain time-frequency representations;
2. Mel filter banks to compress the spectrum onto the Mel scale, aligning with human auditory perception;
3. Logarithmic compression to produce log-Mel spectrograms.

After feature extraction, the resulting acoustic representations are fed into the SqueezeFormer encoder [6]. The encoder consists of a hybrid attention–convolution Transformer architecture [14] that integrates self-attention and convolution modules to capture both long-range dependencies and local acoustic patterns. In addition, the Temporal U-Net [12] design compresses and expands temporal representations, enabling more efficient computation while preserving contextual information, as shown in Figure 3.

Finally, the encoder outputs are passed to a decoding module to generate the predicted transcription. The output representations can be directed to either a CTC decoder [4] or an attention-based decoder for predicting the text token sequence corresponding to the input audio. During training, the model typically employs the Connectionist Temporal Classification loss [4], which facilitates learning mappings between variable-length audio feature sequences and transcript tokens without requiring explicit alignment. The CTC objective [4] computes the total probability across all valid alignment paths between the model outputs and the target label sequence.

4.1.3 Wav2Vec 2.0

Beyond the SqueezeFormer-based architecture [6], we investigate Wav2Vec 2.0 [1], a framework for self-supervised speech representation learning from raw audio. The architecture comprises three primary components: a multi-layer convolutional feature encoder, a Transformer-based context

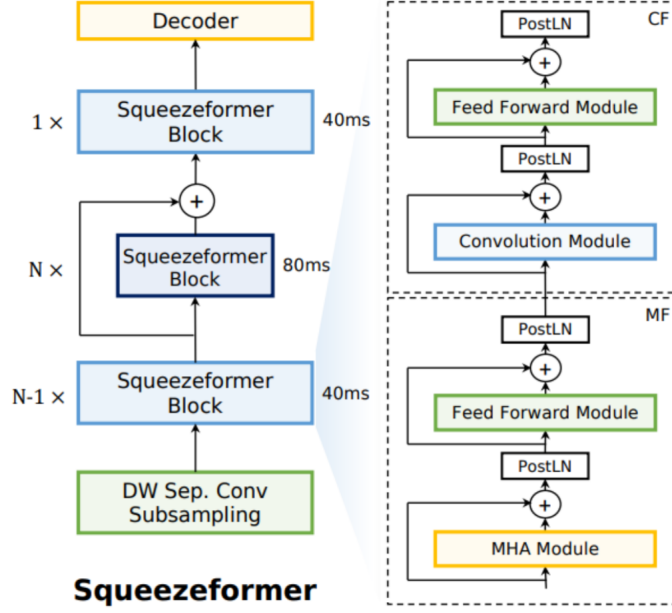


Figure 3: SqueezeFormer Architecture

network [14], and a quantization module. The model is available in two standard configurations: the BASE variant with approximately 95 million parameters (12 Transformer blocks). Wav2Vec 2.0 [1] is first pretrained using a contrastive loss that requires the model to identify the correct quantized latent representation for a masked time step, and is subsequently fine-tuned for ASR using the Connectionist Temporal Classification (CTC) objective [4].

Unlike the SqueezeFormer pipeline [6], which relies on handcrafted features like log-Mel spectrograms, Wav2Vec 2.0 [1] operates directly on raw audio waveforms sampled at 16 kHz. The CNN-based feature encoder consists of seven blocks with temporal convolutions, reducing the input sequence to latent speech representations (z_t) every 20 ms with a receptive field of approximately 25 ms. These representations are then processed by the Transformer encoder [14] to capture long-range dependencies, as illustrated in Figure 4. A key distinction of this framework is the use of product quantization to discretize the continuous encoder outputs into a finite set of speech units during pretraining, which facilitates the learning of robust acoustic phonetics without labeled data.

For ASR downstream tasks, the pretrained context network is augmented with a linear projection layer to predict token probabilities. During inference, transcriptions are typically generated via greedy decoding or integrated with an external n -gram language model via beam search to further refine word error rates (WER). Post-processing steps involve collapsing repeated tokens and removing the CTC [4] blank symbol to produce the final text output.

4.1.4 Qwen ASR 0.6B

In addition to the encoder-centric architectures such as SqueezeFormer [6] and the self-supervised Wav2Vec 2.0 [1], we further explore Qwen ASR [13], a recent large-scale speech recognition model developed by Alibaba Cloud and released during 2025–2026. In this study, we specifically utilize the Qwen3 ASR 0.6B [13] checkpoint.

Unlike the previous models that rely primarily on dedicated acoustic encoders and relatively lightweight decoding modules, Qwen ASR [13] adopts a large language model (LLM)-based, all-in-one architecture. The 0.6B variant contains approximately 0.6 billion parameters and is designed to jointly handle multiple speech-related tasks, including multilingual transcription, language identification, and timestamp prediction. It leverages audio representations derived from the Qwen3-Omni [15] foundation model, enabling tighter integration between acoustic and linguistic features.

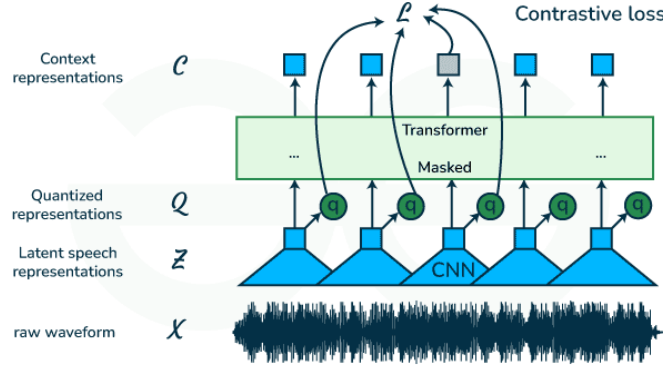


Figure 4: Illustration of Wav2Vec 2.0 framework.

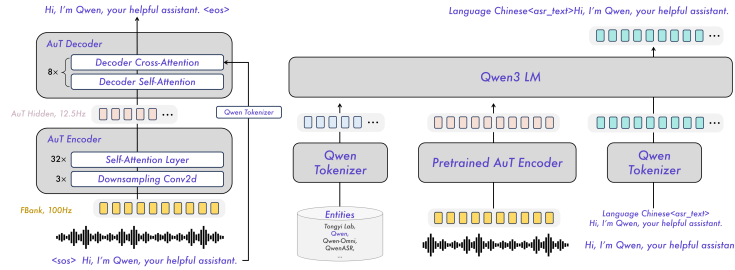


Figure 5: Illustration of QwenASR 0.6B architecture.

Regarding feature extraction, Qwen ASR [13] processes raw audio signals by first converting them into time–frequency representations, typically using log-Mel spectrograms or similar filterbank-based features. These representations capture both temporal dynamics and spectral characteristics of speech. The extracted features are then passed through a neural acoustic encoder—often composed of convolutional subsampling layers followed by Transformer [14] or Conformer [5] blocks—to model long-range dependencies and contextual information. Additionally, leveraging the Qwen3-Omni [15] foundation model enables richer multimodal feature alignment, where audio embeddings are projected into a shared latent space compatible with text representations.

The model architecture, illustrated in Figure 5, adopts an encoder-decoder [14] framework seamlessly integrated with Large Language Model capabilities. The acoustic encoder maps raw audio features into high-level latent representations, while the decoder—leveraging a pre-trained LLM backbone—generates text tokens. Diverging from conventional autoregressive ASR systems, Qwen ASR [13] incorporates Connectionist Temporal Classification [4] alignment heads or parallel decoding strategies to accelerate inference and refine temporal alignment. Cross-attention mechanisms facilitate the bridge between the acoustic encoder and the LLM, enabling a robust fusion of multimodal speech and linguistic context. This unified design allows the model to simultaneously execute speech transcription, language identification (LID), and timestamp prediction within a single forward pass.

For pretraining, Qwen-Audio leverages a massive, multi-task dataset totaling over 653,000 hours of audio data, covering more than 30 distinct tasks and 140+ languages. As detailed in the technical report, the speech-specific component accounts for approximately 453,000 hours, sourced from diverse corpora such as WenetSpeech, Librispeech [10], and Common Voice [9]. This extensive pretraining allows the model to develop a robust “universal” audio representation, encompassing not only supervised speech–text pairs but also capturing acoustic nuances from environmental sounds and music.

Regarding language support, the model provides comprehensive coverage for over 140 languages, including low-resource ones. Vietnamese is integrated into the multilingual framework primarily through the Common Voice [9] and FLEURS datasets. Its performance on Vietnamese ASR is a

result of cross-lingual transfer learning, where the model utilizes shared phonetic representations from high-resource languages to improve recognition accuracy in less-represented ones, even without a primary focus on Vietnamese-specific tuning in the report.

In terms of training strategy, Qwen-Audio employs a multi-stage, multi-task learning framework. Unlike traditional ASR pipelines that might prioritize CTC-based alignment [4], Qwen-Audio utilizes a decoder-only LLM backbone (based on the Qwen series) to perform next-token prediction. The model is trained using hierarchical multi-tasking, where specific “task tags” (e.g., $\langle |ASR| \rangle$, $\langle |LID| \rangle$, $\langle |ST| \rangle$) are used to steer the model towards different objectives like speech recognition, language identification, and timestamp prediction. This unified generative approach eliminates the need for separate non-autoregressive heads, relying instead on the LLM’s inherent reasoning and linguistic capabilities to produce high-fidelity transcriptions and metadata.

4.2 Evaluation Metrics

To comprehensively assess the effectiveness of these baseline models, we employ a set of evaluation metrics that capture both transcription accuracy and computational efficiency. These metrics are divided into two categories: *primary metrics*, which directly measure recognition performance, and *secondary metrics*, which provide additional insights into model efficiency and practicality during deployment.

4.2.1 Primary Metric

- **Word Error Rate (WER):** Word Error Rate is the principal metric used to evaluate transcription accuracy in speech recognition systems. It measures the minimum number of edit operations required to transform the predicted transcription into the reference transcription. Specifically, WER accounts for substitutions, deletions, and insertions at the word level. A lower WER indicates better recognition performance.

$$WER = \frac{S + D + I}{N} \quad (1)$$

where S denotes the number of substitutions, D represents deletions, I indicates insertions, and N is the total number of words in the reference transcription.

4.2.2 Secondary Metrics

- **Inference Time / Sample:** This metric measures the average time required for the model to process a single audio sample during inference. It reflects the computational efficiency of the model and is typically reported in milliseconds or seconds per sample.

5 Experimental Setup

5.1 General Implementation Details

Our experimental framework is organized into two main stages: *Supervised Fine-tuning* and *Inference-based Evaluation*.

1. **Supervised Fine-tuning:** In the first stage, all model architectures, except Qwen ASR 0.6B [13], are fine-tuned using the official training and testing splits of the ViMD dataset [3]. This step aims to adapt the models to the characteristics of Vietnamese regional dialects for accurate evaluation.
2. **Inference-based Evaluation:** After fine-tuning, the fine-tuned architectures and the pre-trained Qwen ASR 0.6B [13] are evaluated through inference on two datasets: the ViMD [3] validation set and the VIVOS [8] corpus.

5.2 Finetune Configuration

The fine-tuning process for all ASR models was conducted using the AdamW optimizer. For the Whisper [11] family, consisting of Tiny, Base, and Small variants, the CrossEntropy loss was utilized, while Wav2Vec [1] and SqueezeFormer [6] were trained using the CTC loss [4].

Table 2: Fine-tuning configuration for all ASR models.

Configuration	Whisper-Tiny	Whisper-Base	Whisper-Small	Wav2Vec 2.0	SqueezeFormer
Optimizer	AdamW	AdamW	AdamW	AdamW	AdamW
Batch Size	16	16	8	4	16
Grad. Accum.	2	2	2	4	2
Warmup Steps	300	300	300	2347	1000
Max Steps	11820	11820	3000	3756	18800
Trainable Params	39M	74M	224M	90M	25M
Loss Function	CrossEntropy	CrossEntropy	CrossEntropy	CTC Loss	CTC Loss
GPUs	T4 x 2	T4 x 2	T4 x 2	T4 x 2	RTX5070Ti

For Wav2Vec 2.0 [1], the CNN feature encoder is frozen during fine-tuning, accounting for only 4.4% of 94.5 million total parameters. The remaining 95.6% comprising the Transformer encoder [14] and CTC [4] projection layer are updated during training.

To address hardware memory constraints, the batch size and gradient accumulation steps were adjusted according to model scale. Whisper-Tiny, Whisper-Base [11], and SqueezeFormer [6] used a batch size of 16 with 2 gradient accumulation steps. For the larger Whisper-Small [11] model, with 224M parameters, the batch size was reduced to 8. Notably, Wav2Vec 2.0 [1] employed a smaller batch size of 4 with a high gradient accumulation of 4 to maintain effective throughput.

Training schedules also varied across architectures. The Whisper models [11] used 300 warmup steps, whereas Wav2Vec 2.0 [1] and SqueezeFormer [6] required 3,756 and 1,000 steps, respectively. The maximum number of training steps ranged from 3,000 for Whisper-Small [11] to 18800 for SqueezeFormer [6]

All experiments were performed on dual NVIDIA T4 GPUs, with the exception of SqueezeFormer [6], which was trained on an RTX 5070 Ti.

6 Results

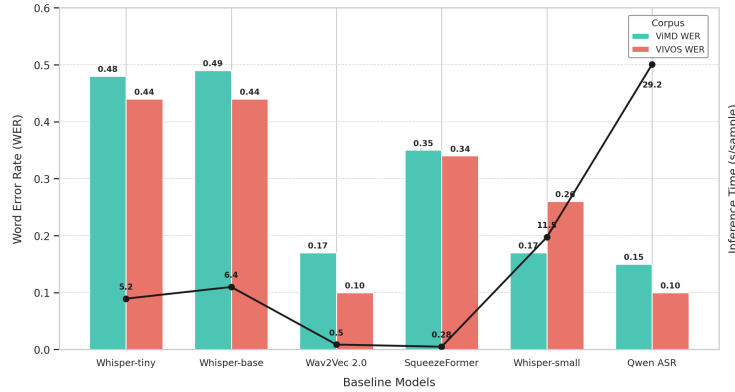


Figure 6: Performance comparison of baseline ASR models after fine-tuning on the ViMD dataset.

6.1 Recognition Performance

Figure 6 illustrates a detailed comparison of the WER across the evaluated models on both datasets. The results reveal significant performance disparities among the architectures.

Starting with the overall best-performing model, Qwen ASR [13] exhibits the highest accuracy on both datasets, achieving a WER of 0.15 on ViMD [3] and a record low of 0.10 on VIVOS [8]. This

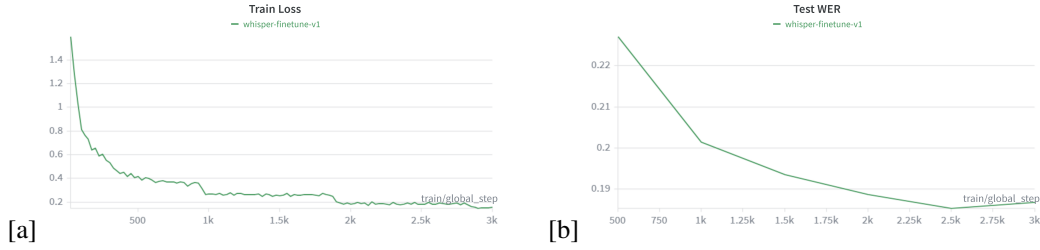


Figure 7: Whisper Small Result

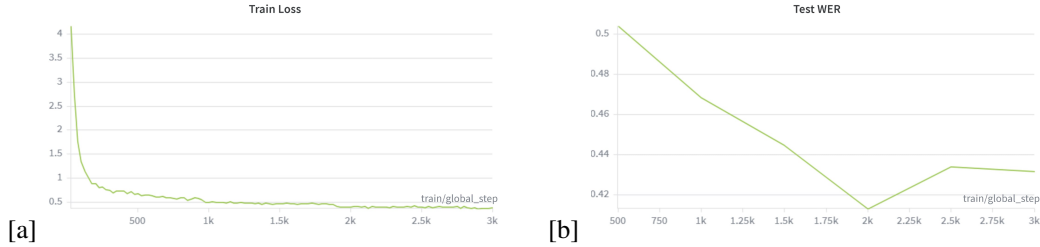


Figure 8: Whisper Base Result

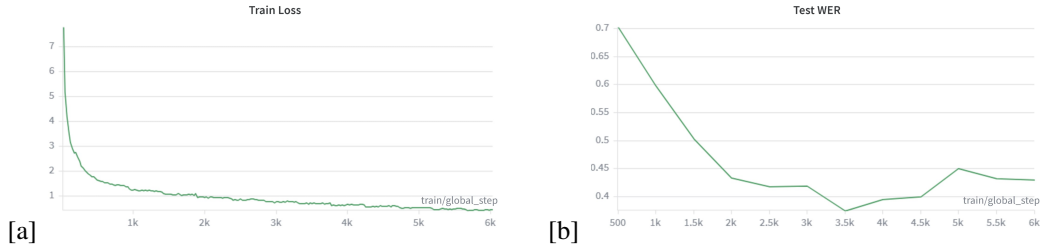


Figure 9: Whisper Tiny Result

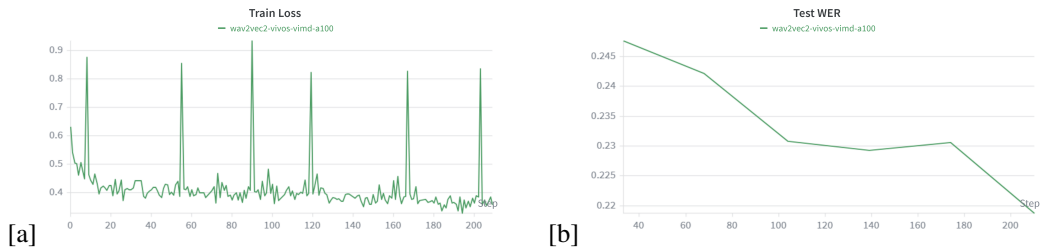


Figure 10: Wav2Vec 2.0 Result

is particularly impressive, as Qwen ASR [13] did not undergo fine-tuning on the ViMD dataset [3], demonstrating the robust Vietnamese language understanding and generalization capabilities of modern large multimodal models.

Focusing on the fine-tuned models, Wav2Vec 2.0 [1] stands out significantly. It achieves a WER of 0.17 on ViMD [3] and 0.10 on VIVOS [8], tying with Qwen ASR [13]. This indicates the architecture’s excellent convergence and its ability to capture local linguistic features when fine-tuned on a Vietnamese dataset.

Looking at model scaling effects, the results demonstrate a clear trend of performance improvement as the parameter scale increases. The smaller variants, including Whisper-tiny and Whisper-base [11], record relatively high WERs, ranging from 0.44 to 0.49. However, scaling up to Whisper-small [11]

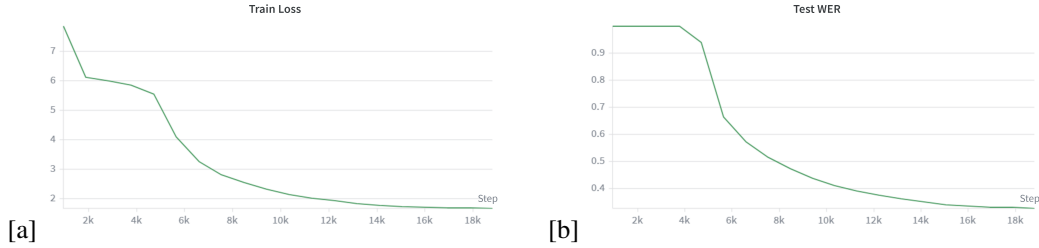


Figure 11: SqueezeFormer Result

drastically reduces the WER on the in-domain dataset to 0.17. Nevertheless, on the out-of-domain VIVOS set [8], Whisper-small [11] still exhibits a moderate error rate of around 0.26.

Across datasets, an interesting observation is that most models record a lower WER on the VIVOS dataset [8] compared to the ViMD [3] validation set. This behavior can likely be attributed to the inherent characteristics of the VIVOS dataset [8], which generally contains more standard utterances, cleaner recording conditions, and less complex vocabulary structures than ViMD [3].

6.2 Computational Efficiency

Beyond raw performance, response time is also a critical constraint when deploying in production environments. The plotted trend line in Figure 6 clearly exposes the trade-off between WER and computational cost.

SqueezeFormer [6] proves superior in raw processing speed, boasting the fastest inference time at merely 0.28 seconds per sample. However, it is Wav2Vec 2.0 [1] that emerges as the most optimal model overall in terms of the performance–speed trade-off. It processes at a highly efficient 0.5 seconds per sample while maintaining an excellent WER, making it exceptionally balanced.

The encoder–decoder architecture of Whisper [11] proves to be severely computationally heavy during inference. The computation time scales significantly with model size, increasing from 5.2 seconds for the tiny version to 11.5 seconds per sample for the small version. Although Whisper-small [11] achieves strong recognition performance, this excessive level of latency makes it impractical for deployment in systems requiring real-time, low-latency responses.

Finally, Qwen ASR 0.6B [13] exhibits a massive computational bottleneck, requiring an unprecedented 29.2 seconds per sample. While it delivers state-of-the-art WER, this extreme latency heavily restricts its utility, rendering it viable only for offline or background batch-processing scenarios where time is not a factor.

7 Conclusion

Based on the empirical results, Wav2Vec 2.0 [1] is proposed as the most viable model for practical deployment due to its exceptional combination of fast inference speed (0.5s) and robust recognition capability (ranging from 0.10 to 0.17 on the WER metric). For tasks strictly unbound by latency constraints but demanding maximum transcription quality, Qwen ASR [13] serves as the optimal choice, though its 29.2s inference time is a critical limiting factor. Models based on the Whisper architecture [11], particularly the larger variants like Whisper-small (11.5s), will require aggressive optimization techniques (e.g., quantization, flash-attention) prior to production deployment due to their significant limitations in sequence generation speed.

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